

# Manuscript Insert: Cap-Released $C^*$ Annulus Certificate

The following text is intended for Paper 2 after author review. It is a standalone draft and does not modify the paper source.

This is a hybrid computer-assisted certificate. The proof text below records the mathematical reduction, the KKT and Hessian margins, the cutoff-zero structure, and the rounding model. The only external data required are the archived cap-1000 candidate, released-face Hessian, theorem-gate ledger, and verification scripts. The exploratory rank-window scans are not part of the certificate.

For a shell-scale vector  $z$  in the finite face below, write  $v_z$  for the corresponding divergence-free Fourier field and set

$$\Phi(z) := \frac{|B(v_z, v_z, \Delta v_z)|}{X(v_z)^2 D(v_z)}.$$

**Theorem 1** (finite one-high annulus certificate). *Let  $\mathcal{F}_{1413}$  denote the finite one-high annulus shell-split face generated by the certified direction cache through the last active shell  $|k|^2 = 2260$ . For the finite-face objective  $\Phi$ ,*

$$\sup_{z \in \mathcal{F}_{1413}} \Phi(z) \leq 0.31226427.$$

*Moreover the certified candidate  $z_*$  satisfies*

$$\Phi(z_*) = 0.31226426322102058.$$

*Proof.* The certificate is carried by three finite ledgers: the cap-released optimizer output, the direct Hessian replay on the released free face, and the theorem gate ledger. The optimizer was rerun with scale cap 1000. Its maximum coordinate is 200.00000000868158, so no coordinate is at the cap. The artificial cap present in the earlier scale-200 certificate is therefore not part of the active proof.

The KKT inequalities are separated into lower-bound, upper-bound, and free coordinates. There are 154 lower-bound coordinates, and the largest lower-bound gradient is

$$-3.5345695459004496 \cdot 10^{-13} < 0.$$

There are no upper-bound coordinates. On the 647 free coordinates,

$$\|\nabla_{\text{free}} \Phi(z_*)\|_{\infty} \leq 3.963733163849133 \cdot 10^{-12}.$$

The active Hessian was recomputed directly at the cap-1000 point using the alias-safe CUDA FFT replay on all 647 free coordinates. Six of these coordinates have zero Hessian diagonal. They are not flat active directions: the cutoff-zero audit verifies that they have zero one-high triads and zero shared one-high triads, with total linear contribution below

$$5.5965044354606178 \cdot 10^{-49}.$$

Thus these six rows are removed by the structural no-triad lemma.

On the remaining 641-dimensional nonzero block, the symmetrized Hessian is strictly negative. At diagonal threshold  $10^{-30}$ , the largest eigenvalue is

$$-3.0235227279516019 \cdot 10^{-23}.$$

Equivalently, after curvature normalization, the negative Hessian has minimum eigenvalue

$$0.69033821039725485$$

and Gershgorin row margin

$$0.50538223782086966.$$

This gives a stable Schur/KKT certificate for the free face.

The residual quadratic gain allowed by the finite-gradient defect is far below the display margin. The candidate is below 0.31226427 by

$$6.778979433352816 \cdot 10^{-9},$$

while the tight residual-plus-target row in the theorem ledger leaves

$$5.9791946083210946 \cdot 10^{-9}$$

after absorbing the active quadratic residual. Standard floating-point rounding is accounted for with

$$\gamma_m = \frac{mu}{1 - mu}, \quad u = 2^{-53}.$$

The denominator replay uses 4838 charged operations, and the remaining post-residual numerator budget allows 172463812 charged operations before the displayed target is exhausted. The finite replay therefore lies below 0.31226427 with the stated rounding model, completing the finite-face certificate.  $\square$

**Proposition 1** (interface with the far tail). *If the far-tail exclusion beyond  $|k|^2 = 2260$  is attached to Theorem 1, then the same displayed value*

$$0.31226427$$

*is the corresponding full one-high annulus certificate.*

*Proof.* The finite theorem closes all shell-split variables through the last shell in the one-high table. The only remaining directions are beyond  $|k|^2 = 2260$ . The far-tail proposition excludes exactly those directions by the separate no-first-variation and corrected-Hessian tail certificate. Hence no admissible one-high annulus perturbation can exceed the finite-face bound.  $\square$

**Certificate artifacts.** The finite theorem uses the cap-1000 optimizer JSON, the cap-1000 released-face Hessian JSON, and the cap-1000 theorem-gate JSON. Full filesystem paths and SHA256 hashes are listed in the accompanying proof packet.

- final candidate: `cap1000_polish`,
- released-face Hessian: `cap1000_free`,
- theorem-gate ledger: `cap1000_verification`.

The recommended archive is intentionally minimal: it contains the three certificate data files and the two verification scripts, not the exploratory search history.